Customer Segmentation

Executive Summary

Context

This project examines data from Kaggle on mall customers demographics including Age, Gender, Income, and Spending Score to create a personalization strategy tailored to each segments preferences.

Segmentation analysis is conducted primarily in Python by applying an unstructured learning technique to identify and segment customers, and transform Age into Generations and Income into Income Buckets.

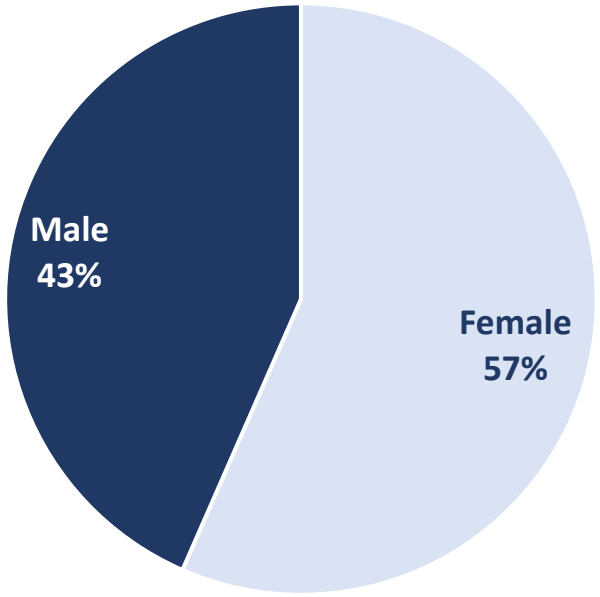
The following slides include insights into Generations, Income Brackets, Gender, and Spending Score with a focus on understanding who customers are, how spending varies across customer demographics, how customers vary by segment, and concludes with key takeaways and recommendations.

**** Analyses and insights are specific to only customers in this study.***

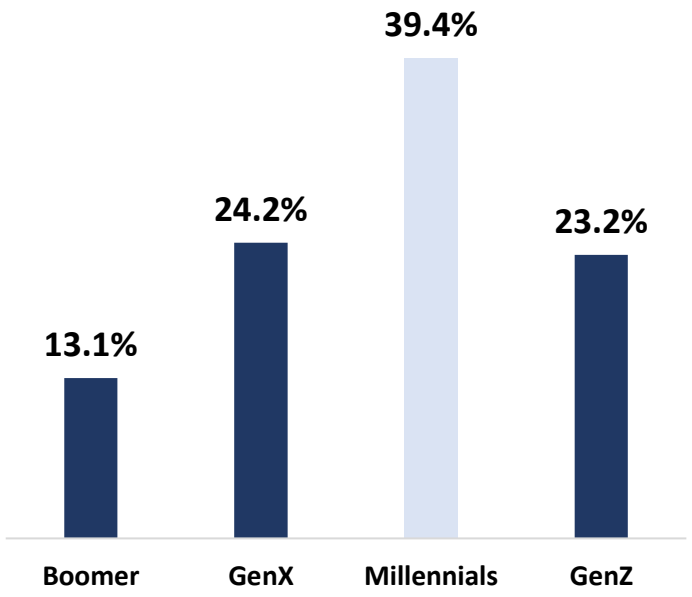
Demographics

Sample skews slightly Female, younger primarily Millennial with more than half of customers 38 years old or less, and in Income Bucket $50K to $74.9K with 50% earning between $42 to $78K per year.

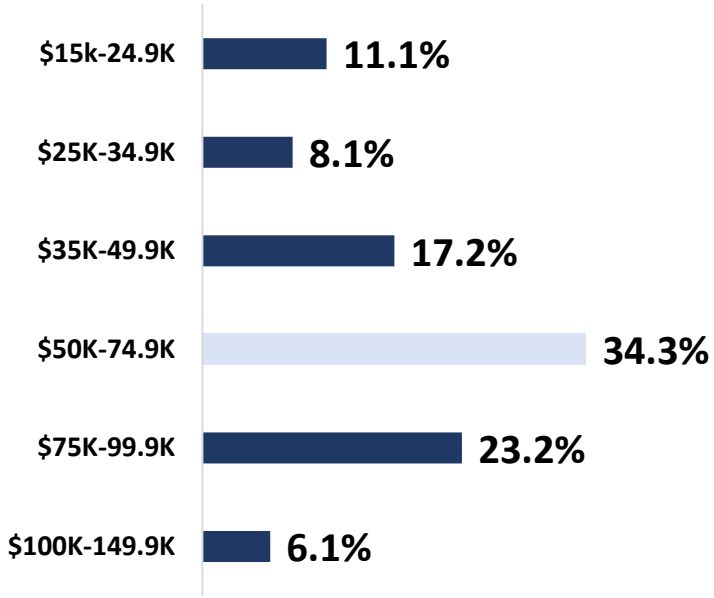
Gender



Generations



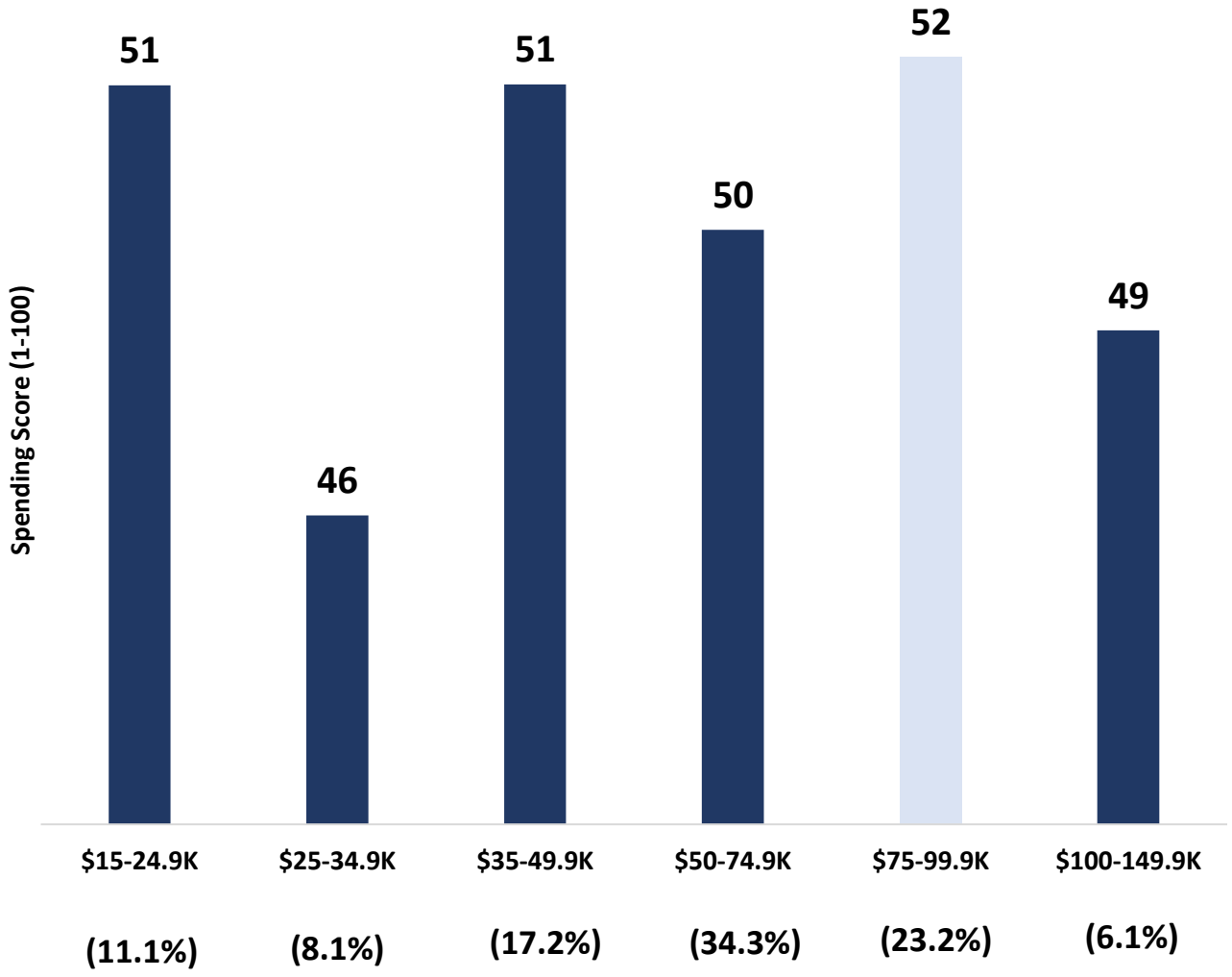
Income Brackets



Spending Scores

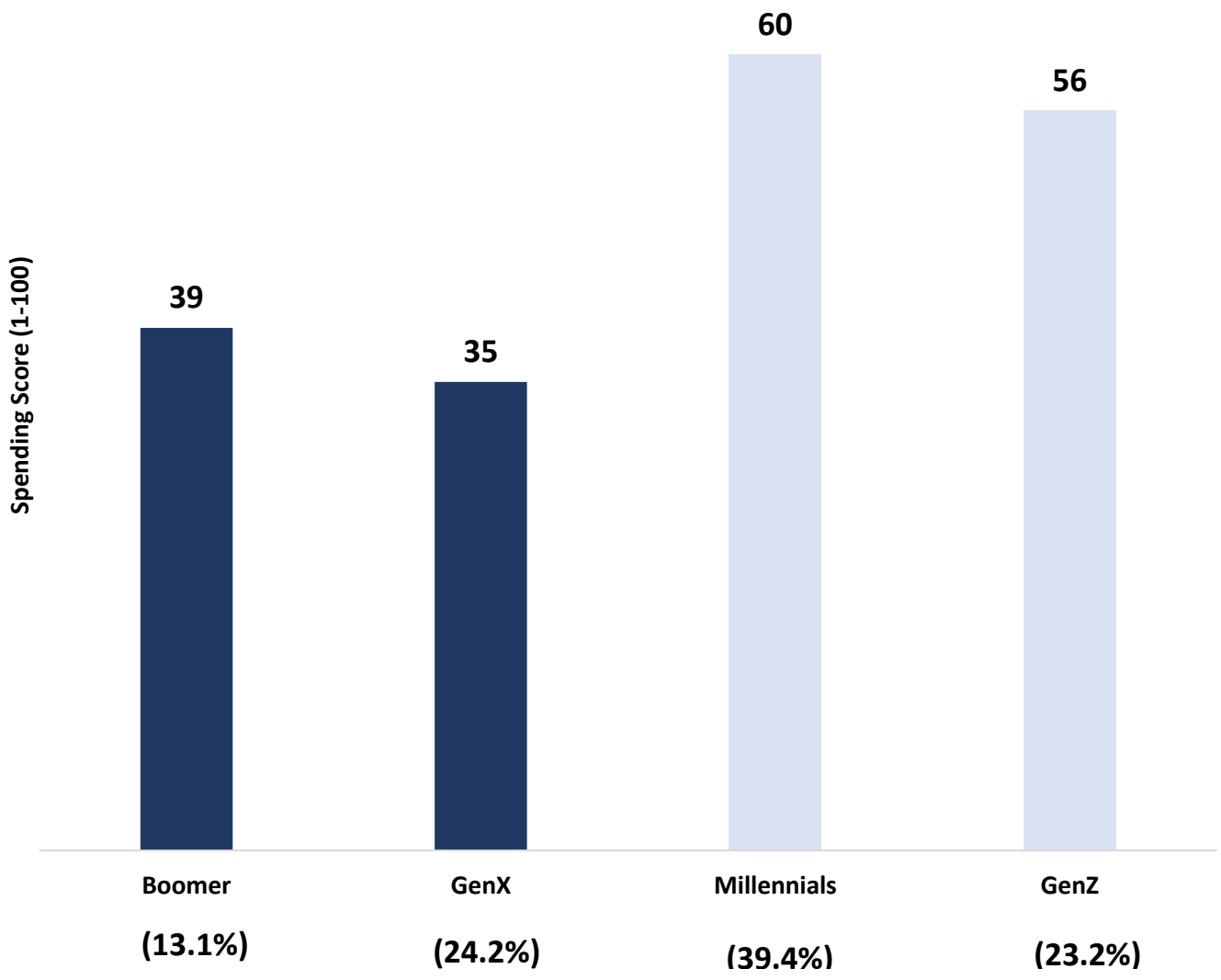
Income Brackets

With a Spending Score of 52-points, Income Bracket $75K-$99.9K comprise just under a quarter of all customers (23.2%).



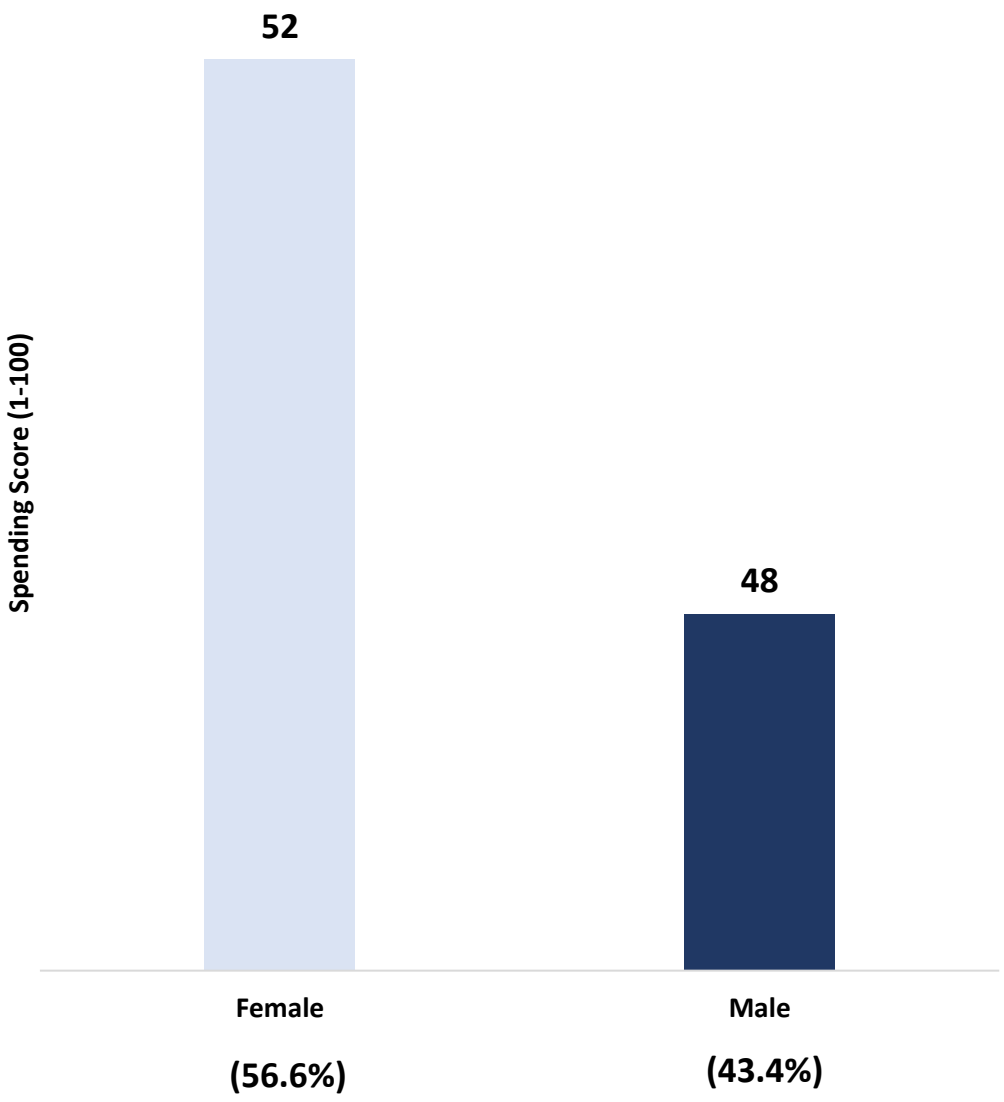
Generations

Millennials & GenZ, make up the majority of sample, 62.6% combined. These two Generations also have highest spending, with an average 58-point blended Spending Score.



Gender

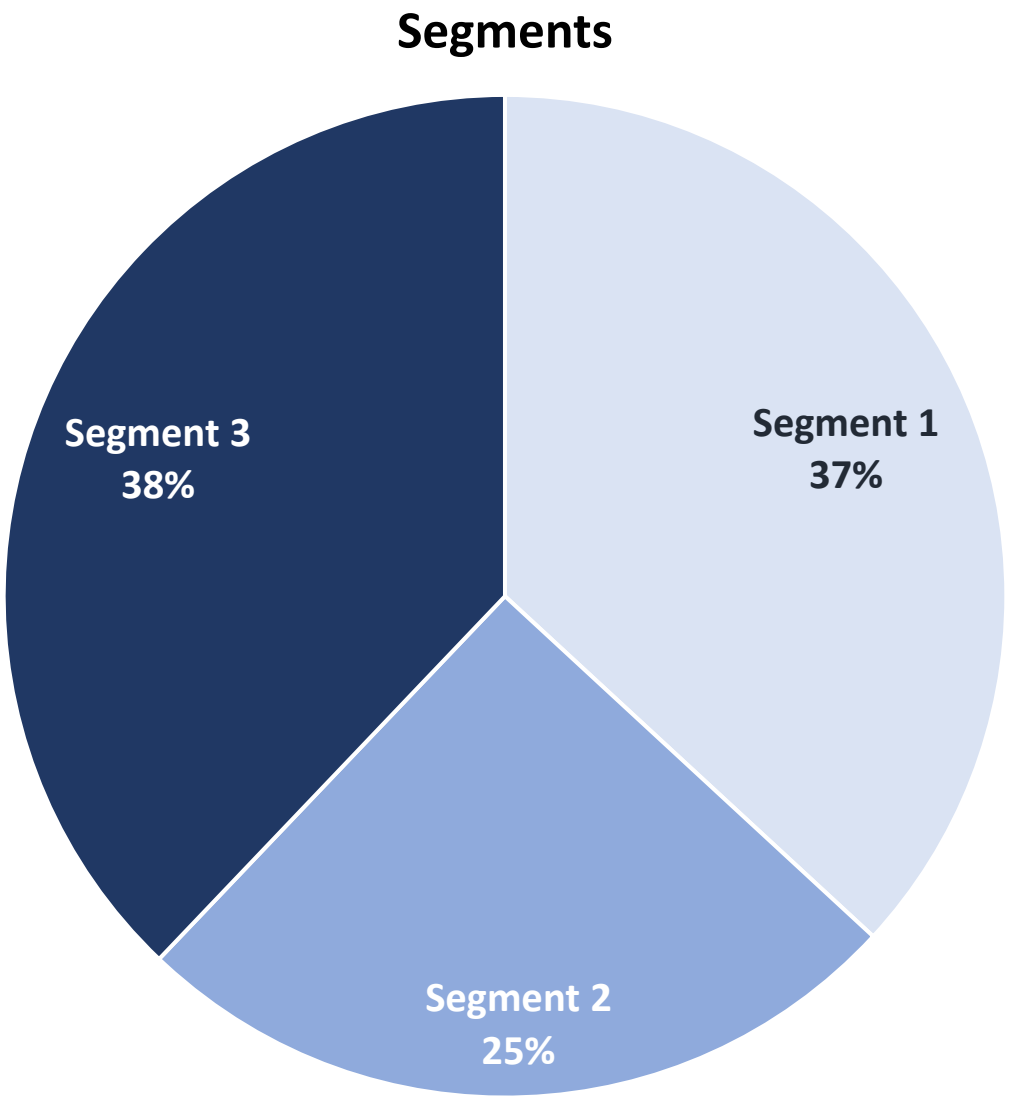
Female Gender comprises just over half of customers, outspending Male Gender with a 4-point higher Spending Score, on average, compared to Male Gender.



Segments

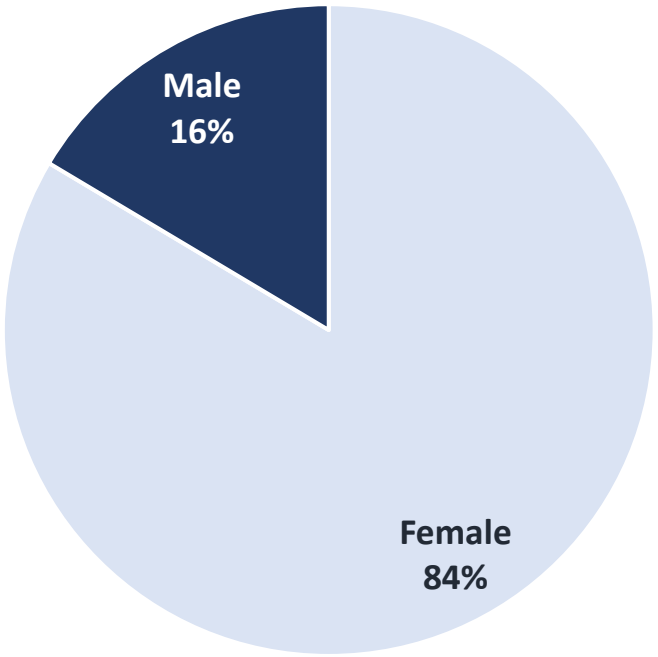
Segments

We divide our sample into three segments based on Age, Gender, and Income.

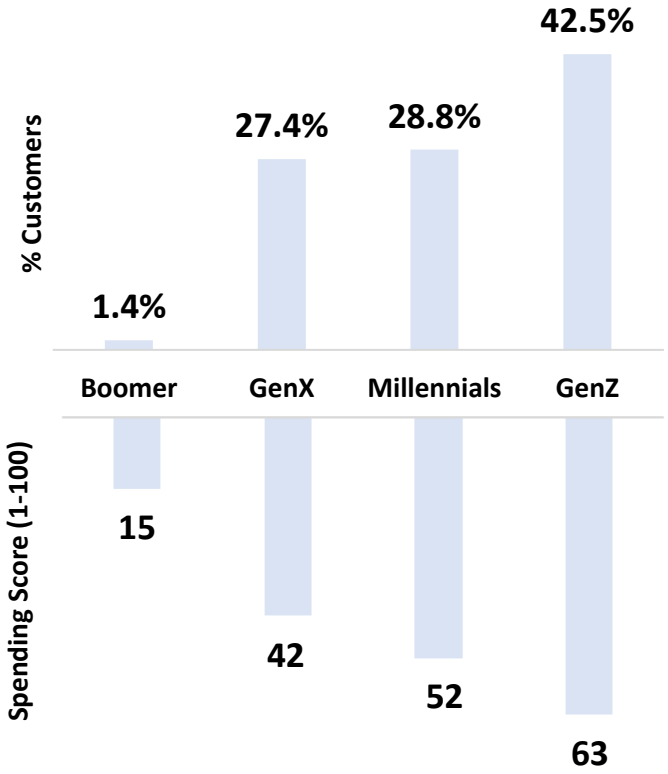


Segment 1 makes up *36.9%* of our sample - comprised primarily of Female Gender, skews younger Boomer, Millennial & GenZ with an average age of 33 years, lower Income Brackets $15K to $74.9K with an average income of $40K, and high-value 53-point blended Spending Score.

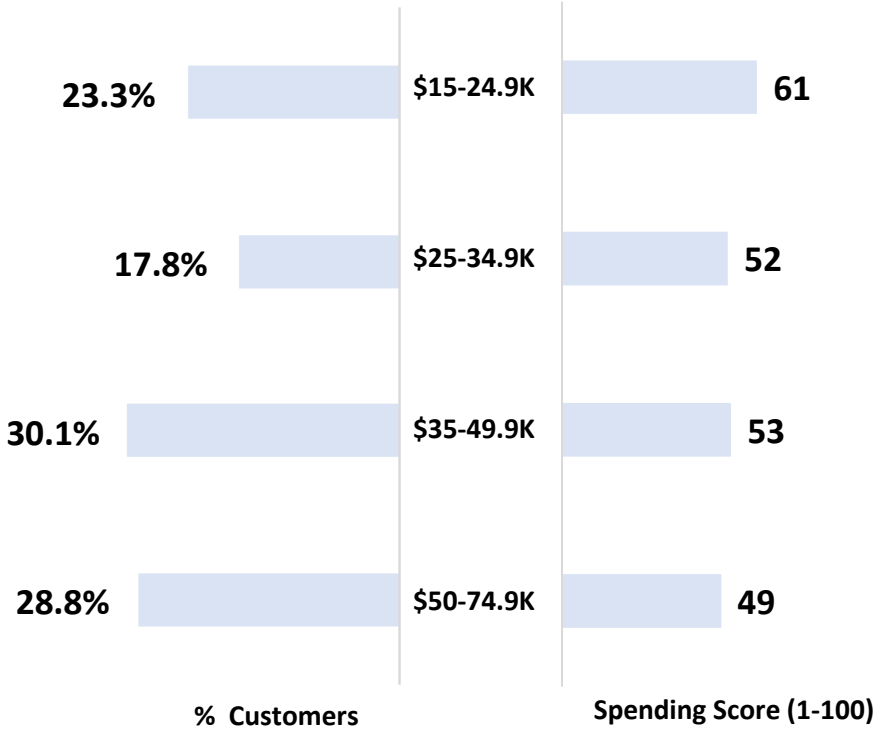
Gender



Generations

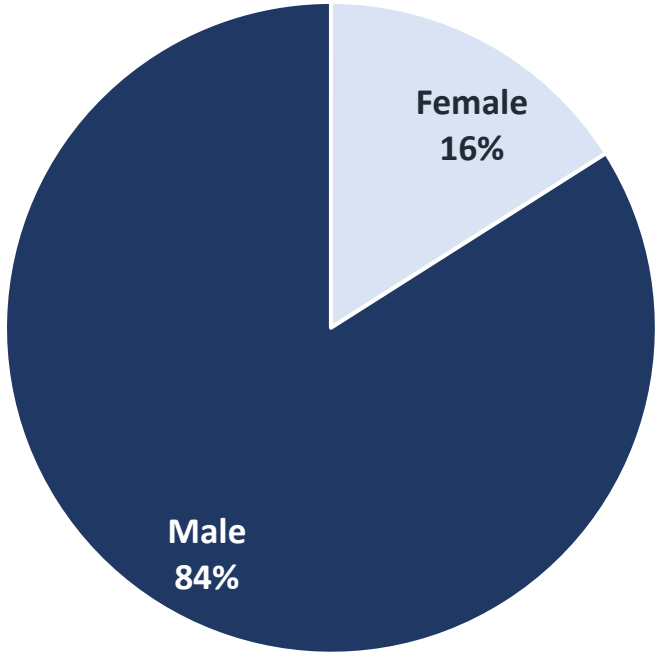


Income Brackets

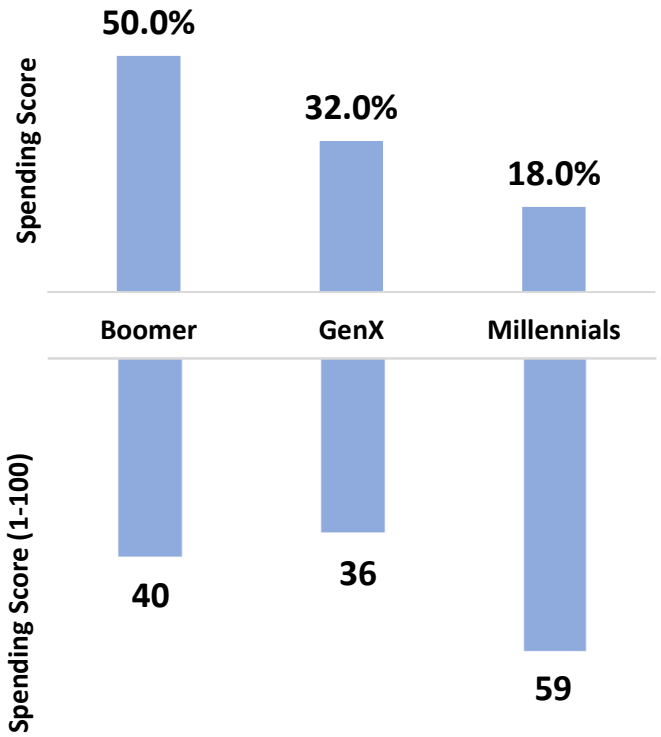


Segment 2 makes up *25.3%* of our sample - comprised primarily of Male Gender, skews older Boomer, GenX, and Millennials with an average age of 55 years, Income Brackets $15K to $75.9K with an average income of $54K, and 42-point blended Spending Score.

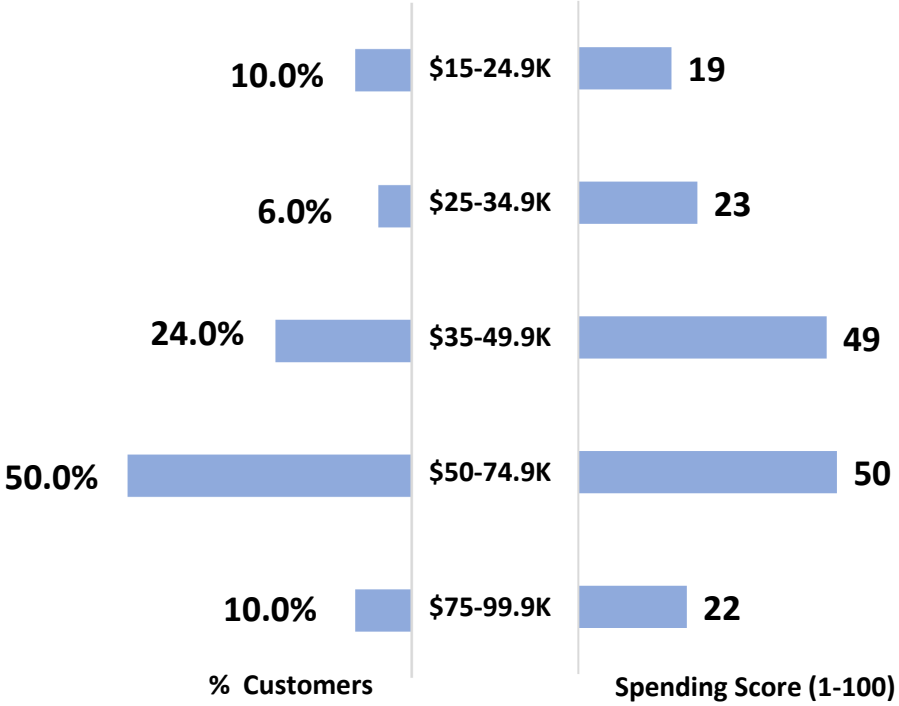
Gender



Generations

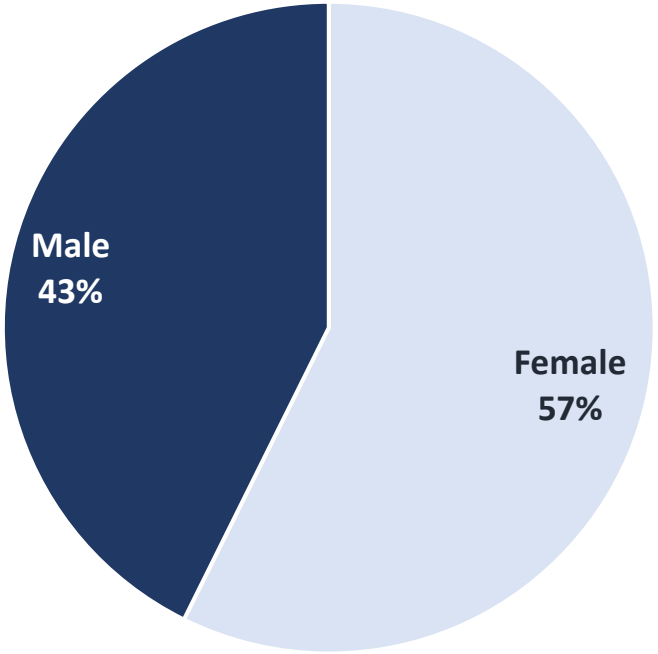


Income Brackets

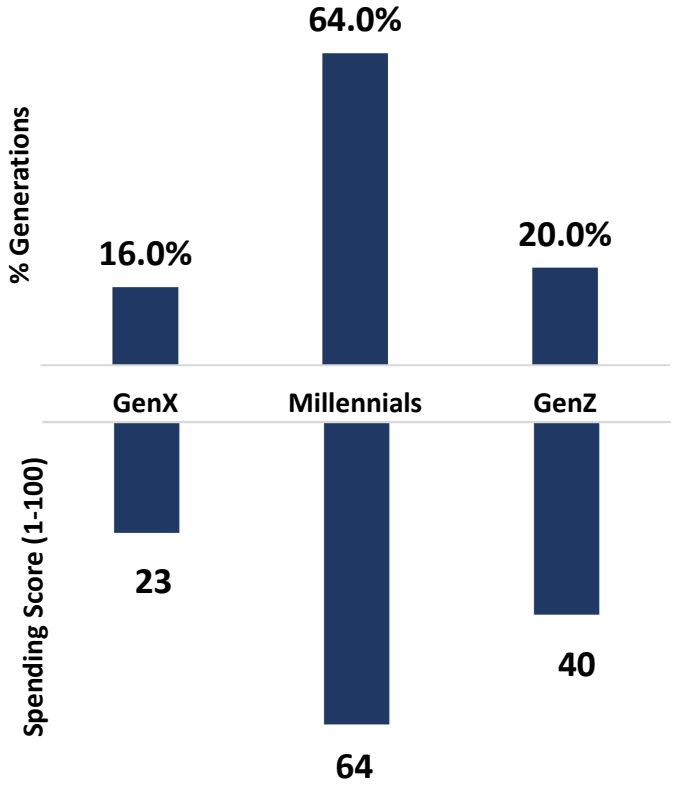


Segment 3 makes up *37.9%* of our sample - comprised of about half Male and Female Gender, skews younger GenX, Millennial, and GenZ with an average age of 34 years, affluent Income Brackets $50K to $149.9K with an average income of $83K, and high-value 52-point blended Spending Score.

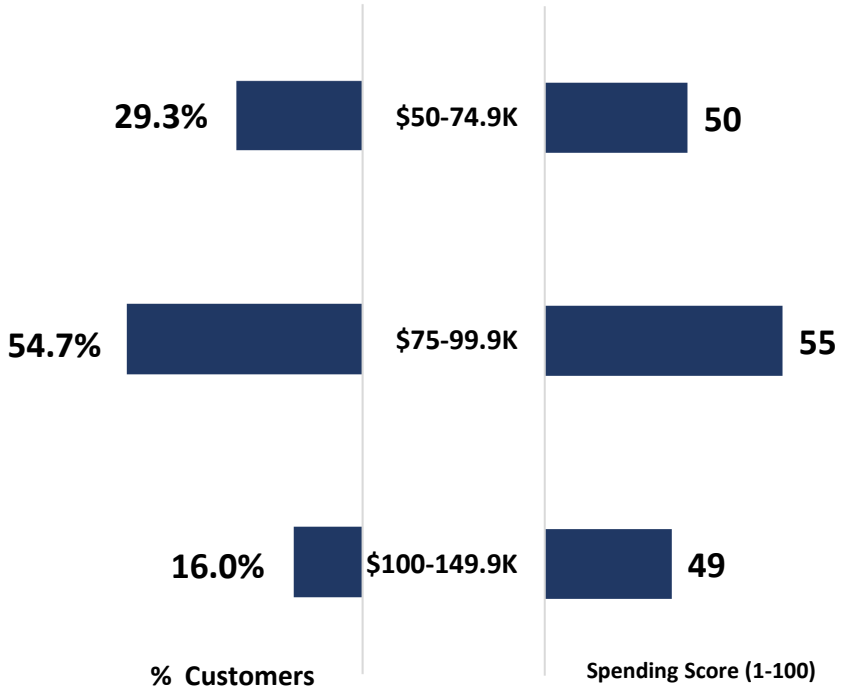
Gender



Generations



Spending Score



Takeaways & Recommendations

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Create a personalization strategy based on customer demographics tailored to each segment:

Segment 1 – primarily of Female, Boomer, Millennial & GenZ, Income Brackets $15K to $74.9K, and 53-point Spending Score.

- *High-value segment:* potential growth opportunity by increasing spend across customers earning $25K+.

Segment 2 – primarily Male, Boomer, GenX, and Millennials, Income Brackets $15K to $75.9K, and 42-point Spending Score.

- Potential opportunity by increasing spend in 26% with Spending Scores less than 49-points.

Segment 3 – Multi-Gender, GenX, Millennial, and GenZ, Income Brackets $50K to $149.9K, and 52 Spending Score.

- *High-value segment:* potential growth opportunity by increasing spend in $50K-$74.9K & $100K-$149.9K.